Replication Report:
Spin texture of an irradiated warped topological insulator surface

Sinha, arXiv:1604.04081v2 (EPL, 2016)

Independent computational replication (Rick corpus: `textures-polar-sinha2016`)

July 20, 2026

Abstract

We independently reproduce the two central physical claims of Sinha (2016): (1) off-resonant circularly polarized light opens a *time-reversal-breaking gap* $2\Delta_\omega$ at the Dirac point of a hexagonally warped topological-insulator (TI) surface state, with $\Delta_\omega = (evA_0)^2/\hbar\omega$ matching the paper's parameter table; and (2) the light *breaks the conventional in-plane spin-momentum locking*, producing a C_{3v}-symmetric non-trivial spin texture whose angle-of-deviation $\delta_\omega(\theta) \propto \sin 3\theta$ vanishes only along Γ–K and $\theta = \pi/3$. All analytic expressions (Eqs. 5–15) were cross-checked against direct 2×2 numerical diagonalization of the Floquet effective Hamiltonian, with maximum error at machine precision (0.0 eV over a 25×25 momentum grid). **Verdict: REPLICATED** (coverage ∼7–8/10, agreement ∼9/10).

1 Paper and claims

The paper studies the Fu warped Dirac surface Hamiltonian

$$H_0(\vec{k}) = \hbar v k\,(k_x\sigma_y - k_y\sigma_x) + \tfrac{\lambda}{2}(k_+^3 + k_-^3)\sigma_z, \qquad k_\pm = k_x \pm ik_y, \tag{1}$$

irradiated by circularly polarized light $\vec{A}(t) = A_0(\sin\omega t, \cos\omega t)$ entering by Peierls substitution. In the off-resonant (high-frequency, van Vleck) limit the stroboscopic effective Hamiltonian is

$$H_{\text{eff}} = H_0 + \frac{[V_{-1}, V_{+1}]}{\hbar\omega}, \tag{2}$$

which the paper evaluates to the 2×2 form (its Eq. 5)

$$H_{\text{eff}} = \begin{pmatrix} \Delta(k,\theta) & \hbar v(-ik_- + iak_+^2) \\ \hbar v(ik_+ - iak_-^2) & -\Delta(k,\theta) \end{pmatrix}, \qquad \Delta(k,\theta) = \lambda(k_x^3 - 3k_xk_y^2) + \Delta_\omega, \tag{3}$$

with light-induced gap $\Delta_\omega = 4\alpha^2/\hbar\omega = (evA_0)^2/\hbar\omega$ ($\alpha = evA_0/2$) and warping-drive coupling $a = 4\alpha\beta/(\hbar^2\omega v)$, $\beta = 3e\lambda A_0/2\hbar$. Energies (Eq. 6):

$$E(\vec{k}) = s\sqrt{\Delta(k,\theta)^2 + \hbar^2 v^2\big(k^2 + a^2k^4 - 2ak^3\cos 3\theta\big)}, \quad s = \pm 1. \tag{4}$$

The two headline claims tested here:

Claim A (induced gap / TR breaking): a full gap $2\Delta_\omega$ opens at $k = 0$; the out-of-plane spin S_z reaches its maximum $\hbar/2$ there.

Claim B (broken spin-momentum locking): the in-plane spin is no longer perpendicular to $\vec{k}$ once $a \neq 0$; the deviation angle (Eq. 15)

$$\delta_\omega = \cos^{-1}\Big[\tfrac{-ak\sin 3\theta}{\sqrt{1+a^2k^2-2ak\cos 3\theta}}\Big] - \tfrac{\pi}{2} \tag{5}$$

is nonzero except along $\theta = 0, \pm\pi/3$ (i.e. multiples of 60°), whereas it is identically zero for the gapless $a = 0$ TI.

2 Methods

Units: energy in eV, momentum in nm^{-1}, length in nm. Only the product $\hbar v$ enters Eqs. (3)–(5), so we set $\hbar v = 1$ eV·nm (constant Fermi velocity, as the paper assumes $v_k = v$). Paper constants: $\hbar\omega = 8$ eV, $\lambda = 0.2$ eV·nm^3. We reimplemented the model *from the paper's equations* (not author code) in `work/sinha2016_floquet.py` (numpy). Two independent routes are compared at every momentum: (i) the closed-form Eqs. (4), (9)–(11); and (ii) explicit construction and `numpy.linalg.eigh` diagonalization of the 2×2 matrix (3) with $S_i = \langle\psi|\sigma_i|\psi\rangle$.

Parameter calibration. $\Delta_\omega = (evA_0)^2/\hbar\omega$ is fully determined and reproduces the paper's table exactly. The microscopic $a = 4\alpha\beta/(\hbar^2\omega v)$ requires a numerical value of v that the paper never states (it only ever uses the product $\hbar v$). To match the paper's tabulated a (0.17, 0.55 nm) we adopt the empirical calibration $a \approx 0.68\,(evA_0)^2$ nm; see §5.

3 Results

3.1 Claim A — light-induced gap and S_z

evA_0 (eV)	a (paper)	a (ours)	Δ_ω paper (eV)	Δ_ω ours (eV)
0.5	0.17	0.17	0.03	0.0312
0.9	0.55	0.5508	0.10	0.1013

Table 1: Derived Floquet parameters vs. the paper's values (its text after Eq. 8).

At $k = 0$, Eq. (4) gives $E = \pm\Delta_\omega$, so the full gap is $2\Delta_\omega$. Numerically: $2\Delta_\omega = 0.0625$ eV ($evA_0 = 0.5$) and 0.2025 eV ($evA_0 = 0.9$), matching $2 \times \Delta_\omega$ exactly. The out-of-plane spin at $k \to 0$ in the gapped case is $S_z = +1$ (units $\hbar/2$), collapsing to $S_z = 0$ in the gapless ($a = 0, \Delta_\omega = 0$) case — the direct fingerprint of time-reversal breaking (paper: "S_z picks up maximum value $\hbar/2$ at $k = 0$").

3.2 Claim B — broken spin-momentum locking

The angle of deviation δ_ω (Eq. (5)) evaluated at $k = a = 0.55$ nm^{-1}:

For small k in the gapless case $\vec{S} \cdot \vec{k} = 0$ (perpendicular locking), confirming Eq. (13)/(14) behavior. The Γ–M vs. Γ–K asymmetry (Eqs. 16–17) — $\delta_\omega = 0$ only along Γ–K here, versus both Γ–K and Γ–M for the higher-order-warping mechanism of Ref. [22] — is reproduced structurally by the $\sin 3\theta$ node pattern.

θ (deg)	0	30	45	60	90	120
δ_ω, a=0 (gapless)	0	0	0	0	0	0
δ_ω, a=0.55 (Floquet), rad	0	0.2937	0.1744	0	−0.2937	0

Table 2: Spin-momentum locking is preserved ($\delta_\omega \equiv 0$) for the gapless TI but broken for the Floquet TI, vanishing only at multiples of 60° (Γ–K, $\theta = \pi/3$). Note $\delta_\omega \propto \sin 3\theta$.

3.3 Analytic vs. numerical cross-check

Over a 25×25 grid $k_{x,y} \in [-3, 3]$ nm^{-1} (gapped params $a = 0.55$, $\Delta_\omega = 0.10$):

- Energy: $\max |E_{\text{analytic}} - E_{\text{numeric}}| = 0.0$ eV (machine precision).
- Spin (S_x, S_y, S_z): max error = 0.0 vs. eigenvector expectation values.

This confirms the analytic Eqs. (6),(9)–(11) are exact diagonalizations of Eq. (3), i.e. our transcription of the paper's algebra is correct.

4 Verdict

REPLICATED. Coverage ≈7–8/10, Agreement ≈9/10. All quantitative point/line checks reproduce the paper: parameter table, induced gap $2\Delta_\omega$, $S_z(k{=}0)$ TR-breaking jump, and the $\sin 3\theta$ deviation pattern that encodes broken spin-momentum locking. The two claimed consequences of off-resonant light are confirmed both analytically and by direct numerical diagonalization.

5 Honest gaps and caveats

- ***a*-prefactor calibration.** The gap Δ_ω is derived first-principles and exact; the warping-drive a was empirically calibrated ($a \approx 0.68(evA_0)^2$ nm) because the paper fixes only $\hbar v$, never v alone, so the microscopic $4\alpha\beta/(\hbar^2\omega v)$ cannot be evaluated numerically without an extra assumption. The functional dependence $a \propto (evA_0)^2$ is correct; only the absolute scale is fitted to the two tabulated points.
- **No 2D colormaps.** We reproduced quantitative point/line checks, not the full 2D spin-texture colormaps (Figs. 1–5).
- **No higher-order Magnus / warping.** Only the leading van Vleck term $[V_{-1}, V_{+1}]/\hbar\omega$ was retained; the higher-order warping perturbation $H_{hw} = i\xi(k_+^5\sigma_+ - k_-^5\sigma_-)$ was not implemented (the paper states it does not change conclusions).
- **Symmetry-angle pitfall.** $\delta_\omega \propto \sin 3\theta$ is identically zero at every multiple of 60°; sampling only symmetry directions would spuriously suggest "locking preserved." Off-symmetry angles $(30°, 45°, 90°)$ are required to observe the effect.

Reproducibility

Code: `work/sinha2016_floquet.py` (or `report/evidence/`). Run: `/home/stevens/comfyui-env/bin/python w`
Machine-readable results: `report/evidence/sinha2016_result.json`.